

CUSTOMER MILESTONE BRIEF

MS #03 - Evidence-Aware Knowledge Graph

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.3.0
Date	2026-08-10
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #3 - Evidence-Aware Knowledge Graph converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Turns a large legacy estate into a traceable map of programs, data, jobs, dependencies, rules, tests, and source evidence.

Why this matters to a customer

Turns a large legacy estate into a traceable map of programs, data, jobs, dependencies, rules, tests, and source evidence.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a deterministic, evidence-aware knowledge graph for the complete pinned AWS CardDemo estate.
- Added COBOL, copybook, JCL, dataset, Java, test, and Maven dependency extraction.
- Added nine curated INTCALC business-rule mappings from legacy evidence to Java and verification.
- Added graph validation, coverage-gap detection, impact analysis, traces, and agent context packages.
- Added implementer/verifier visibility separation for private holdout assets.
- Added content-addressed graph snapshots, source-file hashes, receipts, and JSON Schema.
- Integrated graph regeneration and policy checks into Windows, macOS, Linux, and GitHub CI flows.
- Made snapshot verification compare canonical graph hashes rather than platform-specific gzip headers.

- Added graph architecture, trust model, ontology, query examples, and expansion roadmap.

How it differs from earlier work

MS #3 builds on MS #2 (Verified Java and Spring Batch Candidate). The earlier milestone established its own bounded capability; this milestone adds evidence-aware knowledge graph while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use evidence-aware knowledge graph as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.3.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity